

Ricardo Costa

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PROFILE

Software Engineering student interested in building reliable, scalable software. Strongest interests in static analysis, type systems, compilers, software verification, and tools that improve the way developers code. Also experienced in full-stack web development, cloud computing, and mobile applications.

EXPERIENCE

LASIGE Research Unit

2025 – Present

Researcher

Lisbon, Portugal

Project: Improving the usability of LiquidJava, a refinement type checker for Java, through enhanced verification feedback, IDE support, and overall developer experience

Contributions: 90+ merged pull requests to LiquidJava and 70+ merged pull requests to its VS Code extension

Funding: FCT | LASIGE

EDUCATION

Faculdade de Ciências da Universidade de Lisboa (FCUL)

2024 – Present

MSc in Computer Science and Engineering

Lisbon, Portugal

Current Grade: 17/20

Specialization: Software Engineering

Thesis: “Improving the Usability of LiquidJava”, as a Researcher at the LASIGE Research Unit

Main Courses: Software Design (17/20), Compiling Techniques (18/20), Software Verification and Validation (16/20), Cloud Computing (18/20), Software Security (17/20), Mobile Computing (19/20)

Instituto Superior de Engenharia de Lisboa (ISEL)

2021 – 2024

BSc in Computer Science and Engineering

Lisbon, Portugal

Final Grade: 15/20

Final Project: “NoteSpace: Multi-Platform Web Application for Real-Time Document Collaboration & Sharing” (18/20)

Main Courses: Programming (19/20), Software Laboratory (20/20), Algorithms and Data Structures (16/20), Web Programming (17/20), Mobile Programming (16/20)

TALKS

JNation 2026, Coimbra, Portugal | “LiquidJava: Catch Bugs at Compile Time with Liquid Types” (Co-Speaker)

PUBLICATIONS

“Barista: Synthesizing Typestate Specifications with LLM Agents” | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Accepted at ASE 2026)

POSTERS

“Improving the Usability of LiquidJava” | Ricardo Costa, Catarina Gamboa, Alcides Fonseca

“Barista: Synthesizing Typestate Specifications with LLM Agents” | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Best PhD Student Poster Award)

SKILLS

Programming Languages | Python, Java, JavaScript, TypeScript, Rust, Kotlin, Go, C, C#

Cloud & DevOps | GCP, Cloudflare, Docker, Kubernetes, Git, GitHub Actions, Linux

Web Development | HTML, CSS, React, Node.js, Express, Spring Boot, Vite, Webpack

Mobile Development | Android, Jetpack Compose

Databases | PostgreSQL, Firestore, MongoDB, Elasticsearch

Compilers & Static Analysis | Parsing, Type Systems, Liquid Types, SMT Solvers, LLVM

ACADEMIC PROJECTS

liquidjava [🔗](#) | Contributed to LiquidJava, improving diagnostics and verification

vscode-liquidjava [🔗](#) | Extended IDE support for reading, writing and debugging refinements

NoteSpace [🔗](#) | Real-time collaborative editor with markdown and version control support

aguda-rs [🔗](#) | Compiler for a custom programming language implemented in Rust

Threadit [🔗](#) | Reddit clone REST API built with microservices architecture using gRPC, deployed on GCP

PERSONAL PROJECTS

cf_ai_code_reviewer [🔗](#) | Lightweight AI code review application powered by Cloudflare Workers AI and KV

Sketch.io [🔗](#) | Multiplayer drawing and guessing game using React, Node.js, and Socket.IO

passman [🔗](#) | Secure multi-user CLI password manager with local encrypted storage built in Rust

High-Contrast [🔗](#) | 2D platformer mobile game built in Unity with C#

AdventOfCode [🔗](#) | Solutions for Advent of Code coding puzzles using Python (88 completed)

LANGUAGES

Portuguese | Native **English** | Fluent